

Tomas Benedicto

Head of Payload Architects · Airbus Defence and Space · Toulouse, France
linkedin.com/in/tomasbenedictorinaudo · English, Spanish, French, Dutch · PMP®

I architect satellites end to end.

Aerospace engineer with 10 years in industry, from mechanical design to programme leadership. Today I lead Airbus's **payload architects and mission managers**: we support bids and design the next generation of telecom payloads, across all orbits.

10	10+	8+	5
years in the space industry	satellite programmes contributed to	global operators worked with	languages spoken

PROGRAMMES I'VE WORKED ON

OneSat ✦ Galileo G2 ✦ Galileo ✦ Galileo Thruster Pointing Assembly (TPA) ✦ Deployable Thruster Module Assembly (DTMA) ✦ OneWeb ✦ Eutelsat 172B ✦ SES-14 ✦ Syracuse IV ✦ KMilSat ✦

01 What I bring

Two things set me apart: I've worked at every level of engineering and on every major satellite subsystem.

Every level

Analyst, designer, tester, architect, chief engineer, project manager and now head of the payload architects. That brings perspective, technical expertise, proven ownership, business integration, trust and representation.

Every subsystem

Mechanical, electric propulsion, and RF and digital telecom payload, with expertise in mechanics, materials, electrical, thermal, avionics, propulsion, RF, digital signal processing and end-to-end systems engineering.

Industrial leadership

I draw up the costing and planning of my team's activities, manage its subcontracting and answer for cost, schedule and quality, working with 8+ global operators and key suppliers.

02 Experience

Head of Payload Architects

Airbus Defence and Space · Telecom satellite payloads

Current
Aug 2026 – Present
Toulouse, FR

- Leading the team shaping the future of Airbus telecommunications satellite payloads: building technical excellence, developing people, and delivering competitive solutions that create lasting value for commercial and institutional customers.

Telecom Payload Systems Lead

Airbus Defence and Space · OneSat

Current
Dec 2023 – Present
Toulouse, FR

- Technical lead defining the concept of operations of a flexible digital telecommunications satellite system, leading a team of 10+ payload architects.
- Project manager for the payload, running its industrial contracts, schedule and risk.
- Chief engineer of the OneSat Payload Configuration Manager, spanning space segment, ground segment and in-orbit operations.
- Industrial contracts with global operators (Viasat, SES, Thaicom, JSAT, Optus, SingTel, Eutelsat, Yahsat) and suppliers (GMV, Kratos, TESAT, MDA) under tight cost, schedule and technical constraints.

Satellite Propulsion Lead Engineer & Project Manager

Airbus Defence and Space · Galileo Second Generation and a new European constellation

Jun 2021 – Jun 2024
Toulouse, FR

- Propulsion lead engineer for ESA's next-generation Galileo navigation satellites (Jun 2021 – May 2024): architecture of the electric propulsion subsystem and its mechanisms, verification and validation, and in-flight operations planning, reporting to the satellite chief engineer.
- Project manager (Jun 2021 – Jun 2024): led the end-to-end development of a new plasma propulsion assembly for an innovative European satellite constellation.
- Worked across propulsion, electrical, thermal and avionics engineering; flowed system requirements down to equipment and controlled compliance, deviations and waivers.
- Managed suppliers and chaired critical design reviews and non-conformance boards with international stakeholders; delivered on time, on cost and on quality.

Spacecraft Platform Design & Analysis Engineer

Airbus Defence and Space

Oct 2017 – Jun 2021
Toulouse, FR · Stevenage, UK

- Designed, justified and tested satellite platforms and thermo-mechanical products for telecom satellites in LEO and GEO and for Earth observation satellites: OneWeb, OneSat, Eutelsat 172B, SES-14, Syracuse IV and KMilSat.
- Optimised the design of OneSat's novel structure and of the interfaces between satellites on the launcher and with the launcher itself.
- Co-authored Airbus's international, ECSS-aligned industry standard for modelling and optimising space structures.

Aerospace Engineer, Structural Optimisation
Airbus Defence and Space
Apr 2017 – Sep 2017
Toulouse, FR

- Structural optimisation of spacecraft structures, working to ECSS standards.

Trainee, Structures & Mechanisms Division
European Space Agency · ESTEC
Jul 2015 – Sep 2015
Noordwijk, NL

- In the Spacecraft Dynamics Analysis team, verified the Ariane 5 to satellite interface in preparation for the first Ariane 5 launch of Galileo satellites.

03 Leadership & R&D

2022 – 2024 · Airbus DS
Focal point, nuclear propulsion for space
Airbus's focal point for system-level R&D with nuclear-industry partners toward an ESA demonstrator targeted for 2035.

2018 – 2021 · Airbus DS
Center of Competence board member
Co-authored industry standards for finite element modelling and structural optimisation.

2018 · ESA & Airbus DS
ECSS MET conference paper
Co-author of a scientific paper on design processes for structural parts made by additive layer manufacturing.

04 Education & certifications

M.Sc. Aerospace Mechanics & Avionics
ISAE-SUPAERO · Toulouse
2015 – 2017
Award of Excellence 2017
Specialised in aerospace structures. Research on the failure modes of high-strength fibre composites using infrared techniques. Master students' representative on the university board and student representative of the Royal Aeronautical Society's Toulouse branch.

B.Sc. Aerospace Engineering
TU Delft · Delft
2012 – 2015
Awarded with Distinction

Systems Engineering & Excellence
Airbus DS · Toulouse, Ottobrunn & Getafe
2026
Airbus's chief-engineer track: leadership, system accountability and technical decision-making.

Project Management Professional (PMP®)
PMI
Oct 2024

Architecture & Systems Engineering: Models & Methods to Manage Complex Systems
MIT xPRO · professional certificate
Jan – Jul 2021
The systems engineering framework and MBSE for developing complex systems and managing uncertainty.

05 Toolkit

TECHNICAL
End-to-end systems engineering · Telecom payload · RF · Digital signal processing · Electric propulsion · Electrical · Thermal · Avionics · Mechanics & materials · LEO constellations · AIT · V&V · MBSE · ECSS · Python

LEADERSHIP
Project management · Contract leadership · Negotiation · Facilitation · Strategic thinking · Clear communication

LANGUAGES
Spanish — Native
French — Native
English — Bilingual
Dutch — Professional
German — Elementary

OFF DUTY
⚽ Football · 🏄 Windsurfing · 🎹 Piano